NETHRA RAJENDRAN

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PROFESSIONAL SUMMARY

B.Tech Computer Science (AI) student (2028 batch, CGPA: 8.4/10) with software engineering internship experience and a proven track record of building AI, computer vision, and full-stack systems. Published IEEE research in computer vision and predictive maintenance, with finalist/top-5 finishes across national hackathons. Experienced in Python, FastAPI, React, TypeScript, Next.js, PostgreSQL/MongoDB, Docker, and ML/CV pipelines.

TECHNICAL SKILLS

- **Languages:** Python, C++, C, TypeScript, JavaScript, SQL
- **AI & Machine Learning:** PyTorch, TensorFlow, YOLO, OpenCV, scikit-learn, XGBoost, LightGBM
- **Backend & APIs:** FastAPI, Flask, RESTful APIs, Microservices, AsyncIO, Server-Sent Events (SSE)
- **Frontend Development:** React, Next.js, TypeScript, Tailwind CSS
- **Databases & Storage:** PostgreSQL, MongoDB, MySQL, Qdrant (Vector DB), Schema Design & Indexing
- **Tools & DevOps:** Docker, Git, Linux, CI/CD Workflows
- **Core Computer Science:** Data Structures & Algorithms (DSA), DBMS, Operating Systems, Computer Networks

EDUCATION

Amrita Vishwa Vidyapeetham — Coimbatore, India
B.Tech in Computer Science (Artificial Intelligence) — CGPA: 8.4/10 — *2024 – 2028*

EXPERIENCE

IT Trident – Software Engineering Intern — May 2026 – June 2026

- Architected and deployed 4 FastAPI microservices exposing structured REST API contracts consumed by a TypeScript and Next.js client.
- Implemented an asynchronous hybrid retrieval pipeline combining dense and sparse embeddings with RRF and CrossEncoder on Qdrant.
- Built Server-Sent Events (SSE) streaming and asynchronous request batching to handle analytical workflows without blocking client UI execution.
- Implemented structured logging, request tracing, and input validation to isolate cross-service failures; participated in PR reviews and mentorship.

NX Web – Web Development Intern — Summer 2025

- Developed Flask backend services integrated with modular frontend components, managing the request lifecycle across UI, API, and database layers.
- Encapsulated machine learning inference workflows behind clean REST service interfaces to ensure reliable client-facing execution.
- Debugged issues across unfamiliar production codebases, resolved database query bottlenecks, and collaborated within an Agile workflow using Git/GitHub.

SELECTED PROJECTS

DRISHTI-PAT – AI-Based FSOC Terminal Tracking
Python, Computer Vision, YOLO, OpenCV, ROS2/Gazebo

- Developed an AI-assisted vision pipeline for detecting and tracking remote FSOC terminals and estimating alignment error for coarse pointing.
- Designed a closed-loop architecture linking target detection → tracking → angular-error estimation → terminal gimbal control.
- Built a simulation prototype to evaluate tracking robustness under dynamic platform motion and temporary optical beacon loss.

RoadVision AI – Road Infrastructure Assessment Platform
YOLOv8, LaneNet, Computer Vision, GIS, React

- Built a computer-vision platform combining pothole detection, road encroachment, and lane analysis to generate a quantitative Road Quality Score.
- Developed GIS-based visualization and maintenance prioritization workflows for municipal infrastructure monitoring; National Finalist at MobilityX 2.0.

Integrated License Plate Recognition
YOLOv8, CNN, PyTorch, EasyOCR

- Developed an end-to-end license-plate recognition pipeline integrating YOLOv8 object detection with a custom CNN-based character recognition module.
- Evaluated the architecture against baseline YOLOv8 + OCR models on vehicle datasets, improving character accuracy; published at IEEE ICISC 2025.

ACHIEVEMENTS & LEADERSHIP

- **Smart India Hackathon 2025:** Team Lead for a national-level SIH team; led solution architecture and technical execution.
- **Smart India Hackathon (SIH 2026):** Selected team member & Core Developer for DRISHTI-PAT (ISRO/SAC problem on FSOC terminal tracking).
- **MobilityX 2.0 National Finalist:** Selected as a national finalist in a 24-hour hackathon among 150+ teams; built RoadVision AI.
- **Fiesta Hackathon 2026 Top 5:** Top 5 among 150+ teams for an AI-driven waterborne disease early-warning and surveillance platform.
- **COSMOHACK 1.0 Bounty Winner:** Awarded $25 sponsor reward in the Aptos/Web3 track for EduCred blockchain certificate verification platform.

PUBLICATIONS

- **Integrated License Plate Recognition Using YOLO and CNN for Automated Vehicle Identification**, *IEEE ICISC 2025*, DOI: 10.1109/ICISC65841.2025.11187533.
- **ANN-Based Predictive Maintenance for Mini Grid Black Out Prevention**, *IEEE STPEC 2025*, DOI: 10.1109/STPEC66316.2025.11490377.